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TEM and XAS investigation of fission gas behaviors in U-Mo alloy fuels through ion beam irradiation

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ABSTRACT

In this study, smaller-grained (hundred nano-meter size grain) and larger-grained (micro-meter size grain) U-10Mo specimens have been irradiated (implanted) with 250 keV Xe^+ beam and were in situ characterized by TEM. Xe bubbles were not seen in the specimen after an implantation fluence of 2×10^{20} ions/m^2 at room temperature. Nucleation of Xe bubbles happened during heating of the specimen to a final temperature of 300 °C. By comparing measured Xe bubble statistics, the nucleation and growth behaviors of Xe bubbles were investigated in smaller-grained and larger-grained U-10Mo specimens. A multi-atom kind of nucleation mechanism has been observed in both specimens. X-ray Absorption spectroscopy showed the edge position in the bubbles to be the same as that of Xe gas. The size of Xe bubbles has been shown to be bigger in larger-grained specimens than in smaller-grained specimens at the same implantation conditions.

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1. Introduction

U-Mo metallic alloys have been extensively used for the Reduced Enrichment for Research and Test Reactors (RERTR) program, which is now known as the Office of Material Management and Minimization under the Conversion Program, as research and test reactor fuels [1–5]. These fuels are in forms of either monolithic plates or dispersion plates in which U-Mo fuel particles are embedded in Aluminum cladding. U-Mo alloys have also been proposed as metallic fuel candidates for fast reactors due to their ideal thermal properties [6]. In addition, an ongoing effort is investigating the behavior of these alloys along with U-Zr alloys under fast reactor irradiation conditions [7]. U-Zr binary alloy fuels and U-Pu-Zr ternary alloy fuels exhibit zoned microstructures according to their relatively complicated phase diagrams [8]. In contrast, U-Mo alloy fuels stay in a single irradiation stabilized γ phase. As such, the use of U-Mo alloy fuels have the potential to eliminate some of the issues related to zone formation in U-Zr alloy fuels.

However, there is a significant lack of experimental data of this fuel form under fast reactor operation conditions. An early work by a French team has investigated the behaviors of U-8Mo (8 wt percent Mo) annular fuels under fast neutron irradiations [6]. This is, as far as we know, the only relevant data set available to date. Consequently, in order to qualify U-Mo alloy fuels as fast reactor fuel, systematic assessments need to be carried out. The necessary assessments would need to include investigations on the fission product migration behaviors within the fuel, the chemical compatibility of the fuel with the designated cladding material, the fission gas swelling and release behaviors, etc. These assessments would be unavoidably associated with in-reactor tests which are

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extremely costly. Any out-of-pile experiments shedding light into the above mentioned issues are therefore strongly desired.

Over the years, modeling and simulation have provided powerful assistance to the understanding of physical phenomena. It also supplies a non-costly way to assess and predict the performance of nuclear materials [9]. However, extensive verification and validation efforts are needed before any model can be put into practice. Fission gas behavior in metallic alloy fuels typically capture a lot of attention as metallic fuels tend to exhibit high fission gas release and high volumetric swelling at the same time [10]. However, unlike oxide fuels for which large amount of validation data exist, metallic fuels lack fundamental data in order to formulate reliable models to describe their fission gas behaviors. This is particularly true for U-Mo alloy fuels beyond the RERTR temperature regime (typically lower than 200 °C). Therefore, experimental data for fission gas behavior in U-Mo alloy fuels that could potentially help validate fission gas behavior models are desirable as well.

The high burn-up structure in irradiated UO_2 fuels is characterized by sub-divided grains whose sizes are into the nano-meter regime [11]. Similar to heavily cold-worked metals that exhibit refined grains down to the nano-meter regime, the originally larger-grained (~10 μm) UO_2 fuels, upon prolonged irradiation, suffer from grain subdivision or recrystallization [12]. To the surprise of many researchers, the sub-divided nano-grain microstructure possesses improved properties including better fission gas retention capabilities [13,14]. This has led to the thoughts that nano-grained microstructure may be beneficial to nuclear fuels by better retaining the produced fission gas to reduce the rather significant fission gas release and possibly reduce the normally significant swelling of metallic alloy fuels.

In this work, in-situ TEM heavy ion irradiation was used to investigate the nucleation and growth behavior of Xe bubbles in both smaller-grained (hundreds of nano-meter grain size) and larger-grained (micro-meter grain size) U-10Mo TEM specimens. The aim of the in-situ experiment is three-fold: 1. To provide insight into fission gas behavior in U-Mo fuels beyond the RERTR temperature regime; 2. To establish a useful dataset that can help validate fission gas behavior models for U-Mo alloy fuels; and 3. To shed light into the possible advantages of smaller-grained microstructure in retaining fission gas in nuclear fuels. In addition to the in situ TEM experiment, X-ray Absorption Spectroscopy (XAS) was applied to help characterize the state of Xe in an as-implanted TEM specimen. The aim of the XAS investigation is to clarify the nature of nano-meter sized Xe bubbles in U-Mo metallic alloy fuels.

2. Materials and experimental procedure

The original monolithic U-10Mo thin plates coated with zirconium were received from a group within Argonne National Laboratory (ANL). The alloy was cast into coupon shaped ingots using a graphite mold. The alloy coupon had an approximate dimension of 80.2 mm × 45.6 mm × 3 mm. The carbon content of this alloy coupon was about 1000 ppm. This ingot was first acid cleaned using a mixture of nitric and hydrofluoric acid and then laminated, in a carbon steel can, using pure Zr (99.9% pure) foil with approximate starting thickness of 150 μm, on each surface. The Zr-laminated U-10Mo coupon was pre-heated at 650 °C for 30 min in a furnace, and then co-rolled to a thickness of approximately 0.5 mm. A post-rolling annealing treatment was performed at 650 °C for 45 min. The annealed foil was then removed from the can, and cold-rolled to the final thickness of 0.4 mm. This thickness includes the Zr diffusion barrier layer on each surface. The final thickness of Zr diffusion barrier was approximately 25 μm. The final as-rolled U-Mo foil was then sheared into mini-size foils.

Two TEM specimens were prepared from the mini-size foil through mechanical thinning followed by electro-polishing until perforation. The two TEM specimens were then annealed in situ in a HITACHI 9000 TEM up to 700 °C and the resulting microstructures have been reported in Ref. [15]. Recrystallization processes were observed which drove the originally micro-meter sized bcc U-Mo grains into hundred nano-meter size regime. Oxide and carbide particles have also been observed to form on specimen surfaces.

To investigate the effect of smaller hundred nano-meter size grains on the formation and growth of fission gas bubbles, both TEM specimens were implanted with 250 keV Xe^+ ions. In one experiment, the implantation was conducted at room temperature with a flux of 6.25×10^{15} Xe/m^2 to see if Xe bubbles would form under such condition. Video was recorded during the entire implantation process. The experiment was stopped at intermittent implantation fluence of 5×10^{19}, 7.5×10^{19}, 1×10^{20}, 1.5×10^{20}, and 2×10^{20} Xe/m^2 to take static TEM images and characterize the microstructure. As Xe bubbles were not observed during the implantation process, an annealing was performed in the end to make the Xe bubbles precipitate out. The temperature of the specimen was raised slowly (15 °C/minute) from room temperature to 300 °C and was held at this temperature for 30 min before the specimen was gradually cooled down. The annealing temperature was selected to be 300 °C for two reasons: 1. U-Mo alloys tend to readily oxidize at high temperatures even in a TEM vacuum as good as ~1×10^{-7} bar; 2. Significant thermal grain growth is expected above 400 °C (above the athermal regime of U-10Mo alloy fuel), which would unnecessarily complicate the experiment. In the other experiment, the in situ implantation was conducted at 300 °C. The temperature of the TEM specimen was first raised slowly (15 °C/minute) to 300 °C and the specimen was allowed to equilibrate for 20 min before the ion implantation began. Video was recorded during the entire implantation process. The experiment was stopped at intermittent implantation fluence of 5×10^{18}, 1×10^{19}, 2×10^{19}, 3×10^{19}, 5×10^{19}, 7.5×10^{19}, 1×10^{20}, 1.5×10^{20}, and 2×10^{20} Xe/m^2 to take static TEM images and characterize the microstructure. Much denser data points were taken as Xe bubbles could form at any fluence according to a di-atom nucleation mechanism usually operating in a nuclear fuel being irradiated.

In order to characterize the difference between the bubble formation and growth behaviors between smaller-grained (hundred nano-meter grain size) U-Mo and larger-grained (micro-meter grain size) U-Mo, an additional specimen processed by Focused Ion Beam (FIB) was implanted again with 250 keV Xe ions at 300 °C. A FIBed specimen was used because electro-polished specimens were demonstrated to oxidize significantly at 300 °C whereas FIBed specimens were shown to not oxidize much under the same condition. This may be related to the extensive near surface irradiation damage by Ga ions during the FIB process which somehow protected the specimen from significant oxidation. Similar to the implantation experiment of the electro-polished specimen conducted at 300 °C, temperature was slowly raised (15 °C/minute) to 300 °C and the specimen was allowed to relax for 20 min before implantation began. Video was also recorded during the entire implantation process. The experiment was stopped at intermittent implantation fluence of 1×10^{18}, 2×10^{18}, 5×10^{18}, 7.5×10^{18}, 1×10^{19}, 2×10^{19}, 3×10^{19}, 5×10^{19}, 7.5×10^{19}, and 1×10^{20} Xe/m^2 to take static TEM images and characterize the microstructure.

XAS measurement was performed on one of the electro-polished TEM specimens after the in-situ implantation experiment to investigate the state of Xe bubbles (whether they are solid, liquid or gaseous). XAS is element specific. In this experiment, the Xe K edge (34.561 keV) from the U-10Mo sample was probed with fluorescence detection. XAS spectra show changes based on the electronic structure of the material and the local atomic structure of nearby scatting atoms. Typically used for oxidation state analysis,

fitting XAS spectra has been successfully utilized to estimate pressures in xenon bubbles by fitting the shifts in the oscillation pattern caused by changes in near-neighbor distance associated with changes in pressure [16]. A method has also been established based on poro-elasticity theory to extrapolate the estimated internal gas pressure at low temperature (as measured) to the temperature at which the bubbles form during irradiations [17]. In this extrapolation, the equation of state of xenon at high temperature is borrowed from the work of Casanova [18] and the fuel matrix is assumed to behave elastically.

The MRCAT beamline at ANL in its X-ray Absorption Fine Structure configuration was utilized for this experiment. The 3rd harmonic off of a double crystal Si (111) monochromator was used to access the 34.561 keV Xe K edge, and a combination of absorbers was used to remove higher harmonics from the beam. The Boyd Technology ion chambers filled with a mix of 20% argon and 80% nitrogen was used to monitor the incoming flux, as well as measuring the transmitted beam. The specimen was measured in fluorescence geometry using a 4 channel Vortex detector. XAS measurements for Xe were carried out at room temperature. According to the SRIM [19] estimate of implanted Xe, an average concentration of 2 at. % was obtained. In order to calibrate the measurement, Xe gas was measured first in transmission geometry. The X-ray Absorption Near Edge Spectrometry and Extended X-ray Absorption Fine Structure data were analyzed using IFEFFIT software package with ATHENA and ARTEMIS interfaces for data reduction and EXAFS fitting, respectively [20].

3. Results and discussion

3.1. Nucleation of Xe bubbles upon annealing

Despite a relatively high final fluence of 2×10^{20} Xe/m^2, after careful through focus characterizations, Xe bubbles were not observed in the as-implanted specimen for the smaller-sized grain sample. This is consistent with the results of an earlier work where Xe was also implanted at room temperature into CeO_2 [21]. Moreover, at a Xe fluence of 10^{16}, bubbles are not observed in the as-implanted specimens of many materials (e.g. UO_2 [22] ZrC [23] Si [24]). All these seem to point to quite high a dose threshold for bubble nucleation. The as-implanted specimen was then heated to 300 °C slowly (15 °C/minute) and held at that temperature for 30 min before it was gradually cooled down. Due to the constant drift and slight bending of the specimen, through focus imaging could not be effectively performed during the heating process. The specimen was carefully characterized via through focus TEM imaging after the specimen reached thermal equilibrium. Fig. 1 compares over focused TEM micrographs taken at 300 °C (left TEM micrograph) and at room temperature before annealing (right TEM micrograph). Xe bubbles were clearly seen in the specimen at 300 °C. This is an indication that bubble nucleation happened while the specimen was heated. Average bubble size and bubble areal number density were estimated to be 1.98 nm and 6.27×10^{16} bubbles/m^2, respectively. For the estimate of average bubble size, 200 bubbles from multiple TEM micrographs were measured to provide reasonable statistics.

3.2. Evolution of Xe bubbles in smaller-grained specimen implanted at 300 °C

Effects of ion implantation with 250 keV Xe^+ ion were also investigated at 300 °C. Xe bubbles were not seen until the implantation fluence reached 7.5×10^{19} Xe/m^2. A large number of Xe bubbles were seen simultaneously pointing to a possible nucleation mechanism similar to a spinodal decomposition. This has been explored by Jeff Rest in his work and was referred to as a multi-atom nucleation mechanism [25]. It should be realized that the simultaneous emergence of Xe bubbles could be related to the detection threshold of TEM being reached at that particular point of time. However, measurements of the Xe bubble size suggested a reasonably wide bubble size distribution. Thus, if Xe bubbles were in the specimen from lower implantation fluence and grew in size to become larger than the TEM detection threshold, these bubbles would have been seen to emerge gradually in number instead of emerging simultaneously. In addition, the observed bubbles are sized (1.9 nm in diameter) larger than what the TEM detection threshold would normally be (~1.0–1.4 nm). Consequently, the simultaneous emergence of Xe bubbles does suggest that a multi-atom nucleation mechanism is operating.

Fig. 2 shows, as examples, bright field over-focused TEM micrographs at the fluence of 7.5×10^{19} Xe/m^2 and the final fluence of 2×10^{20} Xe/m^2. Similar TEM micrographs at the fluence of 1×10^{20} and 1.5×10^{20} Xe/m^2 have been taken as well but are not shown here. At each fluence, multiple under-focused and over-focused bright field TEM micrographs were taken from which Xe bubble size and number density were measured. Fig. 3 shows the evolution of measured Xe bubble size and areal number density with respect to implantation fluence. It can be seen that Xe bubble size continues to increase while bubble number density decreases with increasing fluence. The decrease of bubble number density, however, may be partially due to the sputtering induced thinning of the specimen (250 keV Xe ion is estimated to introduce ~10 nm sputtering per 1×10^{20} Xe/m^2 fluence [26]). Some coalescence could have also contributed to the reduction in the bubble areal number density.

3.3. Evolution of Xe bubbles in FIBed specimen (larger-grained) implanted at 300 °C

A number of electro-polished specimens were attempted but all of them got significantly oxidized as to render TEM imaging

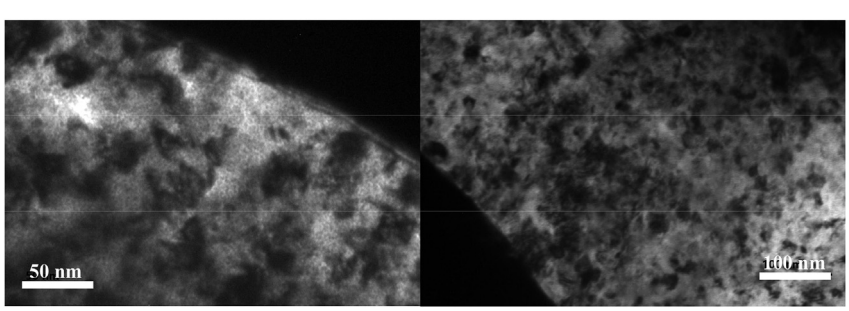

Fig. 1. Over-focused TEM micrographs at ion irradiation fluence of 2×10^{20} Xe/m^2: (left) at 300 °C during annealing and (right) at room temperature before annealing. (All bubbles appear as white dots in under-focused e-beam conditions and black dots in over-focused e-beam conditions).

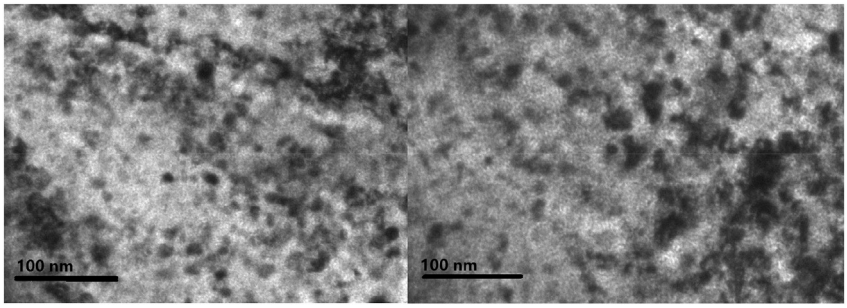

Fig. 2. Bright field over-focused TEM micrographs at the fluence of 7.5 × 10^{19} Xe/m^2 and the final fluence of 2 × 10^{20} Xe/m^2.

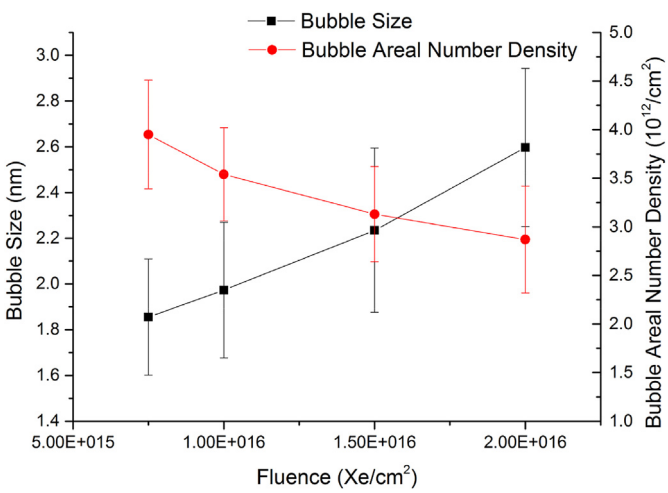

Fig. 3. Evolution of Xe bubble size and bubble areal number density with respect to implantation fluence.

impossible. On the other hand, the FIBed specimen showed minimal oxidation, possibly due to the Ga ion damage on the surfaces of the specimen.

Due to availability of beam time, the implantation fluence only reached 1 × 10^{20} Xe/m^2. To avoid significant loss of Xe to specimen surfaces, the specimen was intentionally kept thicker than normal (usually a FIBed UMo sample is about 30 nm thick considering the high density of the material). This, however, made TEM imaging quite a challenge. Fig. 4 shows a bright field over-focused TEM micrograph taken at the fluence of 1 × 10^{20} Xe/m^2. Xe bubbles can be clearly observed even though the contrast of the micrograph had to be adjusted in order to show these features better. It needs to be pointed out that this fluence was the lowest fluence at which Xe bubbles were observed. However, it does not necessarily mean that nucleation did not happen at any lower fluence. This is due to the difficult imaging condition for through focus examinations. Similar characterization and measurements were conducted on the FIBed specimen to the extent possible (FIBed specimens usually possess less observable thin areas than electro-polished specimens). The average bubble size and bubble areal number density were measured to be 2.89 nm and 1.59 × 10^{16} bubbles/m^2, respectively.

For the purpose of making a good comparison with the smaller-grained specimen, Xe bubble size distributions are compared at the fluence of 1 × 10^{20} Xe/m^2. Fig. 5 shows such comparison. It can be observed that bubble size of the micro-meter grain specimen (FIBed specimen) is slightly larger than that of the nano-meter grain specimen. This is consistent with the results obtained by Spino et al. [12].

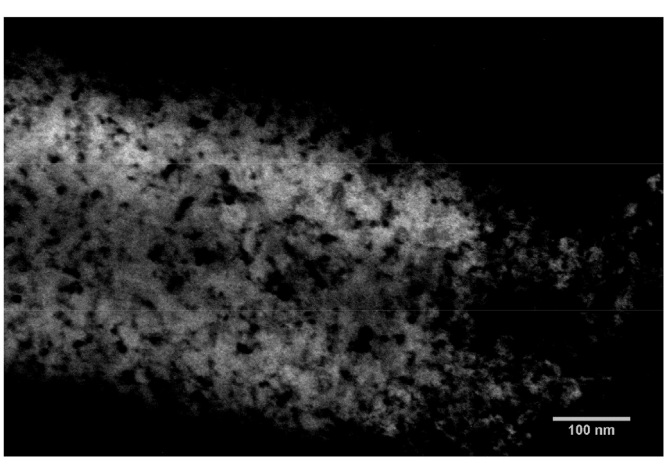

Fig. 4. Bright field over-focused TEM micrograph of FIBed specimen at the fluence of 1 × 10^{20} Xe/m^2.

3.4. XAS measurement of state of Xe

As a necessary calibration step, XAS spectrum of pure Xe gas was measured. A normalized spectrum is shown as the dotted black line in Fig. 6. This spectrum compares reasonably well with existing measurements on Xe gas [27]. A characteristic feature for gaseous state Xe is the hump at around the energy of 34.7 keV [28]. These features are due to multielectron photoexcitation thresholds in the atomic absorption spectrum [27,28]. These features can make structural analysis of the Xe K-edge challenging under the best of circumstances.

The solid, red line in Fig. 6 shows the normalized EXAFS spectrum from one of the annealed TEM specimens (the one on which Xe implantation was conducted at 300 °C). According to SRIM simulation, peak Xe concentration in the TEM specimen was estimated to be at about 4 at.%. However, the implanted Xe atoms were concentrated in a ~50 nm thin layer near the TEM specimen surface due to the relatively low implantation energy used (250 keV). This has led to extremely challenging condition to characterize the Xe K-edge. Grazing incidence was utilized to limit the probe depth to the irradiation area while attempting to miss the implantation layer. Twenty datasets were taken in a total duration of 38 h to acquire the Xe K-edge shown here. The lack of statistics makes it difficult to draw conclusions. However, the edge position is the same in the irradiated material. Also, there is a hint of the multielectron photoexcitation envelop at 34.7 keV in the spectrum from the irradiated U-10Mo sample. This is an indication that Xe atoms are likely residing inside the bubbles in gaseous state.

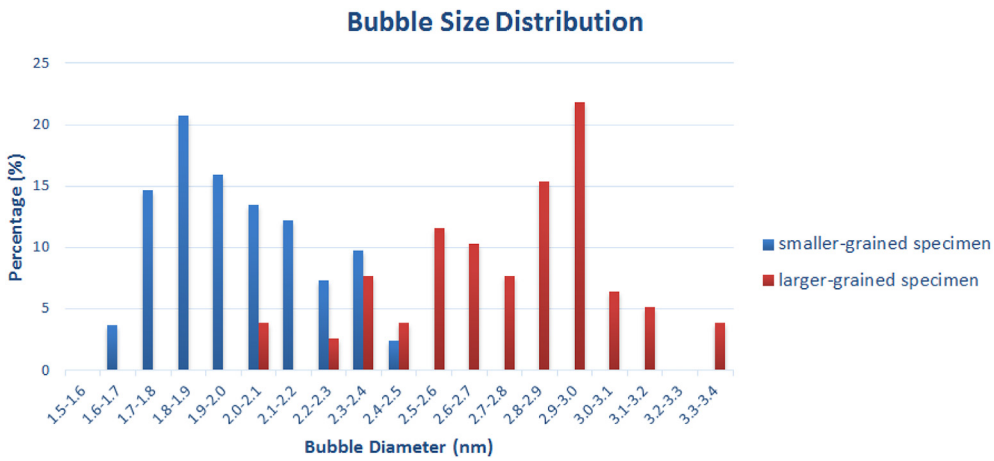

Fig. 5. Comparison of Xe bubble size distribution between smaller-grained specimen and larger-grained specimen at the fluence of 1×10^{20} Xe/m^2.

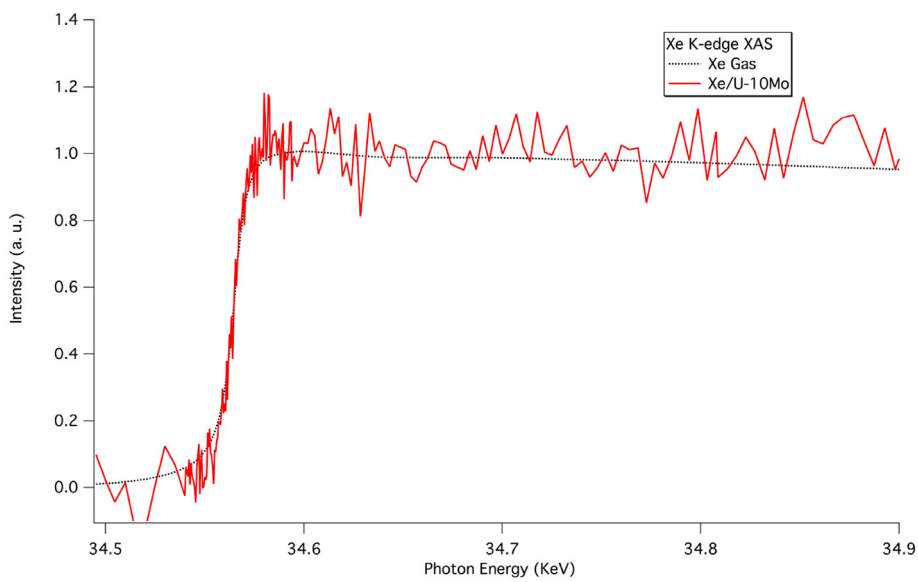

Fig. 6. Normalized EXAFS spectrum of Xe K-edge of a specimen implanted with Xe to the fluence of 2×10^{20} Xe/m^2 at 300 °C and an atmospheric pressure Xe gas.

3.5. Discussion

In both occasions of Xe implantation at 300 °C (on the smaller-grained specimen and on the FIBed larger-grained specimen), simultaneous precipitation of Xe bubbles has been observed. Similar observation has been made in an in-situ experiment where Xe ions were implanted into single crystal Mo [29]. These suggest that a fundamentally different nucleation mechanism, multi-atom nucleation, could be operating, which is quite similar to the Spinodal Decomposition. It is known, however, that researchers tend to model ceramic nuclear fuels (e.g. UO_2) with a di-atom nucleation mechanism. The commonly seen void superlattice in metals and fission gas bubble superlattice in U-Mo metallic alloy fuels are not otherwise observed in ceramic materials. This points to the possibility that the multi-atom nucleation mechanism could be a key differentiating factor for nucleation of bubbles in metallic materials. This mechanism might play an important role in the formation of both void and bubble superlattice.

By comparing the bubble size under very similar implantation conditions (300 °C, 1×10^{20} Xe/m^2 fluence, 250 keV Xe ion beam energy) between smaller-grained and larger-grained U-Mo specimen, it is found that Xe bubbles are larger in the larger-grained specimen (volume ratio of approximately 3:1). In corroboration with the results by Spino et al., this testifies that subdivided nano-grain microstructures tend to put a limiting effect on the size of bubbles [12]. As a result, lower fission gas swelling can be expected in nuclear fuel materials that are smaller-grained.

Through characterization at room temperature, Xe atoms are shown to reside in the smaller-grained U-Mo specimen implanted at 300 °C in gaseous state. Solid state Kr bubbles and Xe bubbles have been observed in ion implanted specimens in the past [30,31]. These solid bubbles were shown to melt upon heating which suggests that gaseous Xe bubbles in our experiment would stay in the gaseous state at high temperatures. Unfortunately, due to the lack of good statistics of the XAS part of the data, conclusions cannot be drawn on the pressure of the Xe bubbles. Without knowing the pressure of the Xe bubbles, Xe atoms lost at the specimen surfaces can only be roughly estimated. If Xe were in solid state inside the bubbles, each Xe bubble with a diameter of 2.597 nm would contain about 400 Xe atoms (assuming a hydrostatic stress of 2.6 GPa) [17]. Thus, total number of Xe atoms in the specimen can be estimated to be 2.29×10^{26} Xe/m^3 assuming solid state Xe bubbles. The total

implanted Xe can be calculated to be 4×10^{21} Xe/cm^3 (at 2×10^{20} Xe/m^2 fluence). Consequently, only 5.7% of the implanted Xe resides in Xe bubbles. If one then takes into account the gaseous state of Xe inside the bubbles, this fraction further drops to below 1% even for highly pressurized gaseous bubbles. While a fraction of implanted Xe could be in solution in the U-Mo matrix, the majority of implanted Xe atoms has to be lost through certain mechanisms. Surface loss due to the thin nature of TEM specimens is well known although only limited quantitative assessment has been made [32]. It is therefore noted that to utilize the experimental results herein for any quantitative modeling, surface loss needs to be assessed very carefully to provide reliable source terms for nucleation and growth of Xe bubbles.

Comparatively, at 1×10^{20} Xe/m^2 fluence, the volume occupied by Xe bubbles in the FIBed larger-grained specimen is about 40% larger than that in the smaller-grained specimen. Through TEM characterization, there is no indication that the Xe atoms are in solid state inside the bubbles (solid state Xe would show an additional family of Bragg reflections in the diffraction pattern. Due to the small nature of the FIBed specimen, similar XAS measurements cannot be performed to determine the exact state of Xe. If one assumes similar bubble pressure between the two scenarios, larger-grained specimen retains more Xe in form of bubbles than the smaller-grained specimen. Smaller-grained microstructure presents large volume fraction of grain boundaries which tend to function as high speed transport tunnels for defects and species such as fission gas atoms. This transport mechanism is believed to have the potential to accelerate the release of Xe to the specimen surfaces in our experiment.

4. Conclusions

In this paper, fission gas behaviors in U-Mo metallic alloy fuels have been investigated by a coordinated effort combining the in situ TEM implantation and XAS. Through in situ TEM characterization, several phenomena were probed including annealing induced nucleation of Xe bubbles and nucleation and growth of Xe bubbles during ion implantation at temperature. Xe bubbles were not observed during Xe implantation at room temperature. However, when temperature was raised from room temperature to 300 °C, Xe bubbles were observed to precipitate. Nucleation of Xe bubbles were observed at around 7.5×10^{19} Xe/m^2 fluence when another smaller-grained U-10Mo specimen was implanted with Xe at 300 °C. When comparing measured Xe bubbles at the same implantation conditions (i.e. ion energy, temperature and fluence), Xe bubble size were seen to be bigger in a FIBed larger-grained U-10Mo specimen than in the smaller-grained specimen. XAS measurements suggested that Xe atoms reside in the bubbles in the gaseous state. Through analyses of Xe bubble size and number density, it was found that significant amount of Xe must have been lost via free surfaces of the thin TEM specimen.

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